Construction Chemicals

SUBMITTAL DOCUMENT

O-Silica

Silica Fume
for High-Performance Concrete



Strength	Permeability	Chlorides	Dosage
Higher	Reduced	Better resistance	5 – 10 %

Project	
Main Contractor	
Consultant	
Submittal No. / Rev.	00
Date	Nov 2026

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METHOD STATEMENT

O-Silica — Silica Fume for High-Performance Concrete

Project Name		**Document No.**	OMX-MS-OSILICA
Main Contractor		**Revision**	01
Consultant		**Date of Issue**	Nov 2026
Product	O-Silica	**Status**	For Approval

1. Purpose

This method statement defines the procedure, sequence, resources, quality checks and safety measures for the supply and application of O-Silica on site. It ensures that the work is carried out safely, in line with the approved drawings and project specification, and achieves the performance stated in the O MIX Technical Data Sheet.

2. Scope of Work

This method statement covers dosing, mixing, placing and quality control of concrete containing O-Silica.

- Approval of the mix design.
- Dosing at the batching plant or on site.
- Mixing and placing of concrete.
- Curing.
- Testing and documentation.

3. Definitions

Term	Meaning
Dosage	Quantity of O-Silica per m³ of concrete or % of cement.
Mix design	Approved concrete proportions.
Slump	Measure of fresh concrete workability.

4. Roles & Responsibilities

Role	Responsibility
Project Manager	Overall responsibility and approvals.
Batching Plant Supervisor	Dosing, mixing and delivery records.
Site Engineer	Placing, compaction and curing.
QA/QC Inspector	Fresh and hardened concrete testing.

Role	Responsibility
HSE Officer	PPE and safe handling.
Storekeeper	Storage and batch control.

5. Materials

Material	Description
O-Silica	25 kg bags.

5.1 Key product data

Property	Typical Value
SiO_2 content	85 – 92 %
Bulk density	500 – 700 kg/m³
Specific surface	15,000 – 25,000 m²/kg
Moisture	≤ 3 %

6. Tools & Equipment

- Calibrated scales or dispensers at the batching plant
- Concrete mixer / truck mixer
- Slump cone
- Cube moulds
- Slump cone, cube / cylinder moulds
- Curing compound or wet hessian

6.1 Personal protective equipment

- Gloves
- Goggles
- Dust mask
- Safety shoes

7. Storage & Handling

- Store dry in closed bags, protected from moisture.
- Store on pallets, off the ground, in a dry, covered and ventilated area.
- Keep containers closed until use; use materials on a first-in, first-out basis.
- Keep containers tightly closed and protect from frost and direct sunlight.
- Handle with care to avoid damage to packaging; do not stack beyond the safe height.

8. Pre-Start Requirements

- Mix design including dosage approved.
- Trial mix completed and approved.
- Dosing equipment calibrated.
- Team briefed.

9. Mix Design & Trial

1. Prepare trial mixes with the approved dosage (5 – 10 % of cement weight).
2. Test workability, strength and other specified properties.
3. Approve the mix design.
4. Record the approved mix design.

10. Dosing

1. Check the batch number of the material and record it.
2. Weigh the dosage for each batch.
3. Add with cement and aggregates.

11. Mixing & Placing

1. Add O-Silica with cement and aggregates.
2. Mix for an extended time as per trial.
3. Place and compact as normal concrete.
4. Work in manageable areas and finish each area in one operation where possible.
5. Check layer thickness and consumption regularly and record the results.

12. Curing & Testing

- Start curing immediately; protect from drying for at least 7 days.
- Protect from sun, wind and rain; keep moist where required during the first 24 hours.
- Barricade the area and display warning signs.

13. Quality Control & Inspection

Check	Stage	Acceptance Criteria	By
Substrate	Before application	Prepared, sound and clean as specified	QA/QC
Climate	During work	Use only in an approved mix design	Site Engineer
Consumption	During application	5 – 10 % of cement weight	QA/QC
Finish	After curing	Uniform, no defects	QA/QC

13.1 Inspection and test plan

Activity	Inspection / Test	Frequency	Record
Material receipt	Product, batch, condition	Each delivery	Material inspection report
Dosing	Quantity per batch	Each batch	Batch ticket
Fresh concrete	Slump, temperature	Each truck	Delivery record
Hardened concrete	Compressive strength	As per specification	Test report

14. Defects & Remedies

Defect	Likely Cause	Remedy
Balling / clumping	Added too fast or at wrong stage	Add gradually with aggregates; extend mixing
Low workability	No dosage adjustment	Adjust superplasticiser in mix design
Plastic shrinkage cracks	Poor curing	Cure immediately

15. Health, Safety & Environment

Hazard	Control Measures
Dust / fibres	Dust mask, gloves, goggles
Machinery	Guards and trained operators
Manual handling	Correct lifting technique

- Wear gloves, safety goggles and a dust mask when mixing.
- Contains cement; alkaline when wet. Avoid skin and eye contact.
- Avoid inhaling dust.
- Eyes: rinse immediately with plenty of clean water for at least 15 minutes and seek medical advice.
- Skin: wash with soap and water; cement is alkaline when wet.

15.1 Waste & environment

- Collect empty containers and residues in designated bins.
- Do not discharge material or washings into drains or soil.
- Allow small quantities of leftover mixed material to cure before disposal; dispose of according to local regulations.

16. Cleaning

- Clean equipment with water.

17. Limitations

- Use only the approved dosage.
- Any deviation from this method statement must be approved by the Site Engineer and the consultant.

Appendix A — Inspection Checklist

Inspection Item	Result	Remarks
Substrate prepared and approved	☐ Yes ☐ No	
Batch numbers recorded	☐ Yes ☐ No	
Material mixed as specified	☐ Yes ☐ No	
Applied at specified consumption	☐ Yes ☐ No	
Curing and protection carried out	☐ Yes ☐ No	
Final inspection passed	☐ Yes ☐ No	

Inspected by	Signature	Date

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Technical Data Sheet

O-Silica — Submittal Document

Construction Chemicals

Product Datasheet
Edition: 01 | Date: Nov 2026
Jointing and Tiling



O-Silica

Silica Fume
for High-Performance Concrete

Product Description

O-Silica is a silica fume (micro-silica) addition for high-performance concrete. It increases compressive and flexural strength, reduces permeability and chloride ingress and improves sulfate resistance.

Uses

- High-performance concrete.
- Precast elements.
- Marine structures.
- Bridges.
- Industrial floors.

Advantages

- Increases strength.
- Reduces permeability.
- Improves chloride and sulfate resistance.
- Dense concrete matrix.
- Dosage 5 – 10 % of cement.

Application

1. **Mix Design**
   Include 5 – 10 % of cement weight in the approved mix design.
2. **Dosing**
   Add with the aggregates and cement at the batching plant.
3. **Mixing**
   Extend mixing time to disperse fully; adjust superplasticiser.

Consumption

5 – 10 %
of cement weight

Packaging

25 kg bags.

Storage & Shelf Life

Store dry in closed bags, protected from moisture.

Technical Data

Typical values at 25 °C

Property	Typical Value	Test Method
SiO_2 content	85 – 92 %	—
Bulk density	500 – 700 kg/m³	—
Specific surface	15,000 – 25,000 m²/kg	—
Moisture	≤ 3 %	—

Consumption

Use	Dosage
High-strength concrete	7 – 10 %
Durability / low permeability	5 – 8 %

Consumption is theoretical and varies with substrate porosity, profile and application method.

Application Conditions

- Use only in an approved mix design.
- Use with a superplasticiser.
- Cure immediately after finishing.

Application Notes

- Use with a superplasticiser.
- Extend mixing time.
- Cure carefully; silica fume concrete is sensitive to early drying.

Health & Safety

- Wear gloves, safety goggles and a dust mask when mixing.
- Contains cement; alkaline when wet. Avoid skin and eye contact.
- Avoid inhaling dust.

First Aid & Cleaning

- Eyes: rinse immediately with plenty of clean water for at least 15 minutes and seek medical advice.
- Skin: wash with soap and water; cement is alkaline when wet.
- Inhalation of dust: move to fresh air.
- Ingestion: rinse mouth and seek medical advice.
- Clean equipment with water.

The information in this datasheet is based on laboratory tests and practical experience and is given in good faith. Users should verify product suitability for their specific application. O MIX reserves the right to update this information without prior notice.

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Test Reports

O-Silica — Submittal Document

To be attached: laboratory test reports supporting the values in the Technical Data Sheet.

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Approvals & Certificates

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To be attached: approvals and certificates, where available.